

(De)Centralization in Blockchain Open-Source Development Community

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Abstract

Open-source blockchain projects have adopted the best practices of open-source software (OSS) governance, like non-profit governance. However, unlike other OSS projects, blockchain projects introduce a token-funding model that creates an additional incentive mechanism. In this study, we focus on how the type of governance and the funding model affect developer participation, rate of innovation, and contributions inequality structure. We find that the developers produce a higher ratio of commits with respect to the following activities (innovation rate) and higher average daily activities (participation) in projects with non-profit governance. Also, the token-associated projects demonstrate a higher level of participation. Nevertheless, we find the inequality of contributions is high across all types of OSS projects. Our findings may help OSS project managers and developers understand the role of governance and the token-funding model in OSS development.

Keywords: open source, governance, blockchain, decentralization.

1. Introduction

Coherent with the idea of blockchain as a decentralized network (Lumineau et al., 2021), the majority of blockchain projects have adopted the open-source model for maintaining their main production process – software development. Unlike traditional vertical organizations, open-source software (OSS) development communities are less prone to have a well-defined hierarchy structure, and their internal governance model is decentralized (de Laat, 2007; Shaikh & Henfridsson, 2017). Moreover, the version control system (VCS) allows the coordination of the distributed effort among autonomous software developers in a similar (decentralized) way as the consensus algorithm allows the coordination of blockchain participants. Though

the OSS literature emphasized the efficiency of the VCS in self-organizing the innovation activity, we make an observation that the leading OSS projects over time have adopted centralized governance features, such as non-profit foundations, e.g., Open Software Foundation or Ethereum Foundation, to further the development and maintenance of technological innovation. As other studies show, decentralization is only efficient to a certain extent in generating continuous software delivery (Gutmans, 2014), and centralized elements also contribute to these processes.

A non-profit is a type of enterprise that provides public good, and it is legally forbidden to pay dividends to anyone involved in it (Hansmann, 1980). Due to their legal status, non-profits typically have centralized corporate governance. Non-profit organizations in the context of OSS are typically called foundations.

Anecdotal evidence suggests that the centralized authority of the non-profit governance board may help resolve conflicts within the community and protect the intellectual property rights of the community in relation to the external environment (O'Mahony, 2003). The efficacy of a non-profit OSS foundation can then be measured by its ability to support development processes and create an environment for sustained innovation. In addition, from a public goods theory perspective, the presence of the external governance body solves the two main problems: coordination of the crowdsourcing effort and sustaining the continuity of effort (Samuelson, 1954). From this perspective, establishing an efficient centralized non-profit for an OSS project may signal the long-term orientation and attract more contributors and contributions to the project.

In addition, unlike non-blockchain OSS projects, blockchain projects are often associated with the cryptocurrency monetization model. This feature differentiates the blockchain OSS projects from other non-profit-governed settings. According to the contract failure theory (Hansmann, 1980), the key premise of the efficiency of non-profit governance

over public goods is a non-distribution constraint, i.e., restriction to the use of the generated returns as managerial compensation. In cryptocurrency-powered blockchains, no contractual obligations restrict the management or developers from purchasing publicly tradable tokens and benefiting from their market performance. Hence, on the one hand, the possibility of participation in returns from their innovative activity may create additional motivations for developers to contribute to the OSS project. On the other hand, the direct developer's participation in returns may create short-term motivations. The deviation from the traditional non-profit theory creates new challenges for non-profit organizations in the OSS context, whose objective is to balance freedom of participation and the long-term sustainability of technology.

Therefore, we aim to explore the effects of the choice of organizational form and the cryptocurrency-monetization model on collaboration in blockchain OSS communities. The role of non-profits in OSS development is underrepresented in the literature, and cryptocurrency rewards are a novel incentive in the OSS innovation model. We aim to fill these gaps by analyzing the development community specifics with and without non-profit presence or token rewards.

Our study also enriches our understanding of the balance of centralization and decentralization elements within decentralized systems, such as blockchain and its applications. Although a big selling point of blockchain is in opening the technological possibilities of decentralization, it is an empirical question of whether decentralization can become an economic reality. Past research has shown that in many aspects, the blockchain ecosystem dynamics is far from a perfectly competitive market (Cong et al., 2021), including the mining market (Arnosti & Weinberg, 2022), validator market (Benhaim et al., 2021), block creation market (e.g., MEV Boost), etc. Despite excellent casual discussions of the development process of blockchain projects (e.g., in Narayanan et al., 2016), we are unaware of any systematic data-driven research into the blockchain developer community dynamics. Our study aims to cover this gap by providing empirical evidence of OSS collaboration in blockchain-related products.

2. Theoretical background

Our study draws from the OSS literature to explore the evolution of the process of collaboration within the developer's network. An open-source form of organization is often associated with a private-public innovation development model (von Hippel & von Krogh, 2003). Open-source organizations benefit

from the decentralized network of voluntary contributors, who create a public good and yield private benefits in the collaboration process (Ye & Kishida, 2003). Oftentimes, these networks lack formalized hierarchy but rather rely on intrinsic and extrinsic motivations to contribute. Motivation to learn spurs contributions, and contributing to the community increases the learning effect (Lave & Wegner, 1991). In addition, reputation and status benefits motivate developers to sustain their contributions in the long term (Jergensen, 2011; Nakakoji et al., 2002).

Nevertheless, the free-riding problem persists in the OSS communities. The ability of individuals to free ride on the efforts of others without contributing in return. In more traditional centralized organizations, e.g., firms, contractual obligations and formal hierarchical authority help to alleviate the free-riding problem. However, such controls are absent in the voluntary networks of the OSS developers (Wasko et al., 2004). Despite the free-riding problem, prior OSS literature finds other mechanisms to incentivize participation, such as reputation in the community (Lerner & Tirole, 2005). The presence of the organizational addendum in the form of a non-profit may strengthen the reputational mechanism due to the presence of external management. On the other hand, the presence of the centralized element may also weaken the individual motivations to contribute and disrupt the private-collective innovation process.

The continuous contribution is the primary objective of the OSS project (Grewal et al., 2006). Among the powerful extrinsic motives are the potential to generate private rewards through providing services related to the focal OSS (Lerner & Tirole, 2005), improving skillsets, or adding the project to the portfolio (Lakhani & Wolf, 2005). Unlike the classic software development communities, participation in blockchain-related communities is often associated with direct token rewards (Meng et al., 2023), and the development effort creates tangible market value in the form of the token's market price. The presence of a tradable token and the market feedback about the platform improvements span beyond developers' learning and reputation motivations. The type of OSS projects with market value is new and has not been addressed in the prior literature. We aim to raise this point in our study.

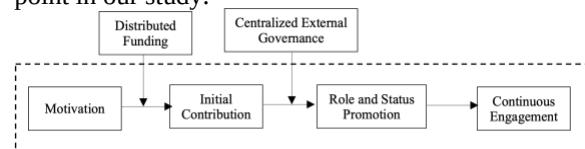


Figure 1. Conceptual model of the long-term OSS sustainability

Figure 1 depicts the conceptual model of our investigation. We suggest two factors that may affect the OSS collaboration: (1) the *organizational model* and (2) the *funding model*. Both factors mediate the two main problems in the public good domain: coordination and funding. First, the governance structure may introduce a hierarchy that will affect the developer's ability to grow in their role and, hence, impede some of their motivations to continue contributing to the OSS project. Second, the direct capitalization of the developers' efforts through the token market can create additional competition among peripheral developers who will produce multiple competing versions of the software. To manage the inflow of innovation, developers should have sufficient time and attention resources to review the proposed changes. The increased competition and demand for the reviewers' resources may affect the long-term OSS development collaboration.

3. Research methodology

We use a quantitative approach to analyze the open-source development collaboration. First, we present descriptive data on the non-profits within major OSS projects. We document similarities and differences in OSS project organization and funding that help us identify the direction for empirical tests. Next, we run the statistical analysis of the OSS contributions in the largest development communities, including blockchain-based ones. We also use the network analysis to show the network structure and the socio-economic analysis to illustrate the inequalities in contributions. Lastly, we use regression analysis to estimate the long-term OSS development of blockchain-based projects.

We use the GitHub data for our empirical analysis – one of the most popular platforms among developers across the world (GitHub, 2024). GitHub provides over 20 event types, ranging from new commits and fork events to opening new issue tickets and commenting on them. To gather all these GitHub activities, we take advantage of the GitHub Archive hosted on the Google BigQuery platform (Hoffa, 2016). We focus on activities most used by software developers to assess project quality and make decisions on whether to join the open-source project (Dabbish et al., 2012), namely commits, pull requests, issues, forks, and watches. We also distinguish between the core and high-effort development activities (commits) and peripheral (all others). We refer to developers who perform each of these core and peripheral activities, respectively.

4. Analysis

4.1. Non-profit OSS governance model

The main premise of the OSS literature suggests that the OSS projects are formed voluntarily through a distributed network of software developers (von Hippel & von Krogh, 2003). Formal hierarchical control is absent in such distributed collaborative networks as it prevents spontaneous work and emergent innovations (Yamauchi et al., 2000). Nevertheless, when exploring the organizational aspects of the largest OSS projects, we find a striking commonality in adopting the centralized coordination model through a non-profit organization. We summarize the results in Table 1.

The comparison of projects in Table 1 suggests that the presence of the non-profit foundation as a satellite of the development community is a common feature among large OSS projects. Non-profit foundations grew their presence despite their centralized nature, which contradicts the hacker ethos of OSS. The examples of such a model are provided by O'Mahony (2003), and the governance model classification is best described by De Noni et al. (2013) and characterized as open-source-based. The first non-profit foundations started appearing at the beginning of the 2000s ("dot-com" boom) when the demand for software was increasing, and many companies requested the OSS software. The centralized entity, such as the foundation, became the centralized agent in the software distribution chain (O'Mahony, 2005).

Table 1. Popular OSS projects' foundations

Project	Year of First Commit	Foundation Establishment Year	# of repositories	Funding Sources (D=Donation, C=Crowdfunding, I=Income)
Python	1989	2001	81	D,I
Linux Kernel	1991	2000	1	D
Apache	1995	1999	2,500	D,I
Mozilla	1998	2003	2,400	D,I
Kubernetes	2014	2015	78	D,I
Ethereum	2013	2014	400	D,C

As Table 1 shows, the average time from the first line of code written to the non-profit foundation has shortened from 10 years on average in the 2000s to less than two years in the 2010s. Such reduction indicates recognizing the non-profit organizational model's effectiveness for coordinating innovative activity. Also, the emergence of centralized non-profit organizations in many well-established OSS projects illustrates the importance of institutionalized leadership (Selznick, 1984).

Nevertheless, the difference in the development timeline of the different projects does not give us a compelling answer as to whether the blockchain-based

projects are to follow the same evolutionary path as other OSS projects. Hence, we continue our analysis using empirical data from the set of contemporary OSS projects.

4.2. Effect of governance and funding models on OSS community productivity

We extend our analysis to the other OSS projects following our theoretical framework. Table 2 presents the classification of OSS projects within the two suggested dimensions: the presence of the external non-profit organization and token-funding availability. Though the prior OSS studies have explored the different governance models (de Laat, 2007; De Noni et al., 2013), there is little evidence of how the diverse models affect the innovative activity of the OSS community. We use the OSS development data to address this gap in the literature.

Table 2. OSS project governance matrix

		Funding	
		No Token	Token
Governance	No Foundation	e.g., Vue.js	e.g., Bitcoin
	Foundation	e.g., Hyperledger	e.g., Ethereum

Note: “No Foundation” projects do not have non-profit or for-profit organizations associated with them; “Foundation” projects have non-profit organizations associated with them.

Consistent with the theory of public goods, we expect that the innovative development effort will be higher under the non-profit foundation model. Hansmann (1980) suggests that in the presence of the non-distribution constraint, the non-profit management will invest more effort in improving the quality of the public good. In our case, the effort can be measured by the new code added to the software through commits. We measure the extensivity of the contributions as the ratio of high-effort code changes (commits) over the number of low-effort activities (forks and watches) and label this measure as the *rate of innovation*. The rate of innovation allows us to implicitly capture the ratio of productive activities relative to free-riding activities.

In addition, free-floating tokens may intensify the extrinsic motivations for developers and create additional stimulus to contribute. Prior studies confirm that extrinsic rewards are an important mechanism to encourage participation in crowdsourcing (Boudreau et al., 2011; Jeppesen & Lakhani, 2010). In our case, the extrinsic motivation of the monetary reward from token ownership may increase the participation rate of both core and peripheral developers. We measure the *participation rate* as the average daily number of activities per developer. The participation rate allows

us to capture work intensity by contributing developers. Figure 2 illustrates the difference in the OSS project productivity depending on the governance and funding models.

As we see from Figure 2, projects associated with the non-profit organization (e.g., Ethereum and Hyperledger) have a higher level of innovation. This result is aligned with the public goods theory assumption that the non-profit organization is an efficient manager of the public goods (Hansmann, 1980). In addition, the management of the non-profit organization will exert an effort to motivate the community to produce innovation, e.g., by sponsoring developers and increasing their visibility in the community. We also observe that the public token-associated projects (e.g., Bitcoin and Ethereum) demonstrate a higher level of community participation. This result is linked to the extrinsic motivation of token appreciation rewards. Developers are important stakeholders of the OSS software, and tokens allow them to effectively capture the value of their efforts.

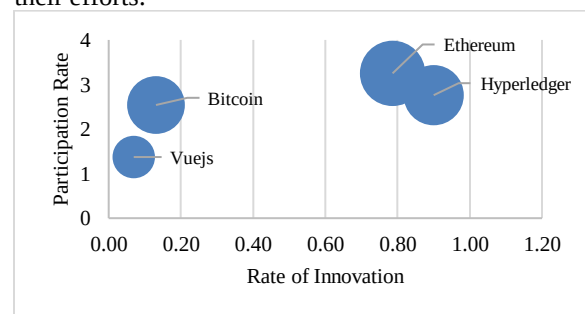


Figure 2. Effect of the governance and token on OSS productivity

Following the theoretical predictions, we find that the non-profit structure and token funding model present an opportunity to affect the OSS development as a public good. To demonstrate the generalizability of our findings, we plan to extend our sample of OSS projects within each of the indicated categories and verify our findings using the cross-sectional regression analysis.

4.3. Centralization inside the OSS communities

The productivity outcomes seen in the previous section may be a result of different effort distribution structures. In this section, we further analyze the productivity of developers broken down by their roles and contributions (Jensen & Scacchi, 2007). Such breakdown analysis is important as (1) the burden of development within a small group of developers may stall production and increase the risks for other projects that use the OSS, and (2) a small group of

developers may be a symptom of constraints that could affect the motivations of developers. Moreover, the open-source, decentralized model frequently serves as a means to reach power equity. Therefore, the natural step is to analyze the distribution of contributions among developer roles and identify whether there is a tendency for some groups of developers to dominate. Figures 3 and 4 illustrate the structure of developers and their respective activities. Figure 3 depicts the total number of activities and shares of each GitHub platform activity.

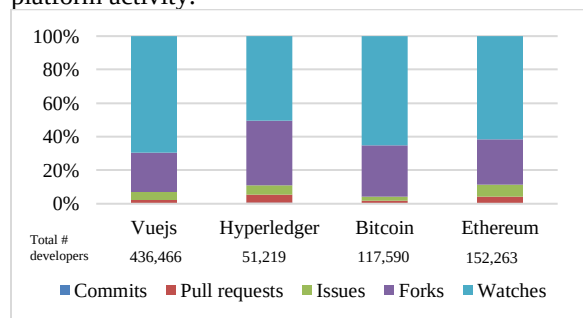


Figure 3. Distribution of developers by roles in OSS communities

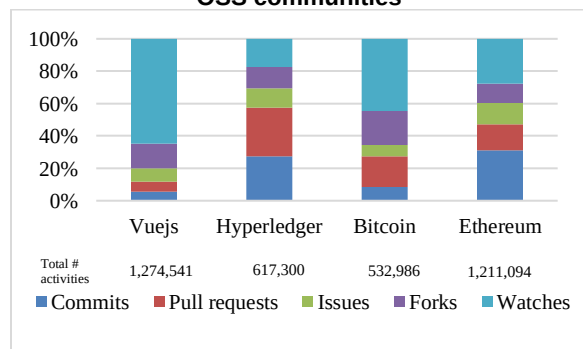


Figure 4. Distribution of activities by types in OSS communities

First, we point out that a low percentage (<1%) of OSS developers across all four projects commit code to the main branch, i.e., make modifications that affect the functionality of the software. This percentage is higher in the projects associated with the foundation model – above 0.1% of total developers. Moreover, less than 20% of developers contribute to the code in the broad sense (commit, pull request, or create an issue). Based on prior research, these types of activity signal the OSS quality (Dabbish et al., 2012; Petryk et al., 2023). Hence, the software quality and its performance are maintained by the active minority. Conversely, around 80% of the community are passive observers, which, however, signals the popularity of the project and its visibility in the global OSS community. Nevertheless, such a high concentration of activities among the small cohort of contributors is common among OSS projects. For example, Ye and Kishida (2003) show that 99% of people who use

Apache are passive users, i.e., do not contribute new code to the software.

The results in Figures 3 and 4 largely show the pattern of unequal distribution of effort within OSS communities independently from the governance and the funding model. Thus, this suggests the natural centralization of effort within open-source organizations. On the one hand, centralization may be a sign of efficiency and trust (Buskens, 1998), thus allowing the core minority to be the gatekeepers of the valuable software. On the other hand, centralization may be viewed as a sign of power (Dosi & Marengo, 2015). In this context, power is not necessarily viewed as destructive, but rather, it can help coordinate activities and achieve efficiency. Nevertheless, the *perception* of power and centralization in the OSS community may play an adverse role in the development.

Following Buskens (1998), we also use network analysis to show the network structure and estimate the possible trust relationships. We use the network density, modularity, and clustering coefficient measures to evaluate the decentralization and modularization of work in OSS communities. The detailed network analysis results are available in Petryk and Li (2023).

4.3.1. Contribution inequalities. To quantify the level of centralization, we use the Gini coefficient and Lorenz curve to estimate the contribution (in)equality level in the developers' community. The Gini coefficient measures statistical dispersion intended to represent inequality (Gini, 1936). For example, the Gini coefficient for the number of activities performed by Ethereum project developers in 2013-2022 is 0.844, which indicates a rather large concentration of activity among the top developer population by their activity.

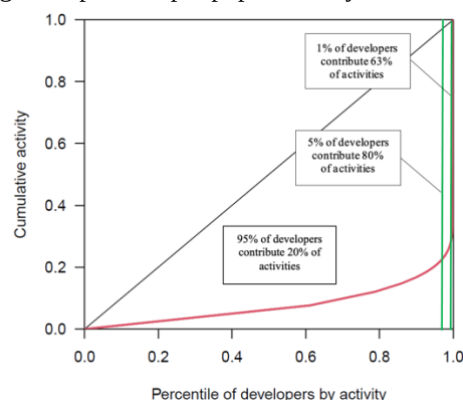


Figure 5. Lorenz curve of Ethereum developer activities

The Gini coefficient is usually calculated based on the Lorenz curve, which plots the proportion of the population's total income (y-axis) cumulatively earned by the bottom percent of the population. For

example, by calculating the Lorenz curve for the Ethereum developers' contribution to the codebase, receive the curve depicted in Figure 5. The horizontal axis represents the percentage of developers, and the vertical axis is the cumulative percentage of activities contributed to the Ethereum codebase over 2013-2023.

The Lorenz curve shape in Figure 5 suggests higher contribution inequality among GitHub Ethereum developers. The curve lies below the diagonal line of equity. Such behavior suggests that over 80% of all activities are contributed by around 5% of developers, while the rest (95%) of developers contribute less than 20%. These numbers are consistent with the prior research. For example, Mockus et al. (2000) report the top 3.8% of developers contributed more than 83% of the modification requests (commits) to another known OSS project - Apache. The author also reports that the inequality is less severe in commercial (closed source) software – between 46% and 77% of commits are performed by 14% of hired developers.

Based on the estimates in Figure 6, the structure of contributions is similar across the three OSS projects. As we observe from our analysis, the bulk of productive development activity is concentrated among the top 5% of developers across both non-profit or no-non-profit governed and token-funded or no-token-funded communities. One explanation for this tendency is that some level of centralization alleviates conflicts and helps manage innovation activities in one direction. Furthermore, such a finding suggests the applicability of the prior OSS theory in predicting behavior in token-funded communities. Therefore, we test if some of the prior findings hold using the example of one of the largest blockchain communities.

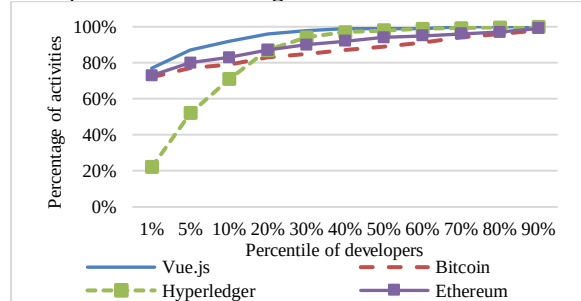


Figure 6. Cumulative distribution of activities among developers in the OSS communities

4.4. Ethereum Hypothesis Testing

To confirm the previously received evidence that major OSS theoretical findings hold for blockchain-based projects, we further test several hypotheses to shed light on Ethereum's blockchain development dynamics. Ethereum has one of the largest blockchain development communities and provides an important

technology for the whole Web 3 infrastructure (Shin, 2022). Building on the existing OSS literature, we test the four hypotheses to assess how the existing developer network dynamics affect the (i) innovation and (ii) diffusion of the project.

Prior OSS literature has established the importance of community diversity for project awareness and diffusion (e.g., Medappa & Srivastava, 2019; Setia et al., 2012). However, none of the prior studies have looked into blockchain projects. Given that a recent survey of blockchain developers (Bosu et al., 2019) shows the difference between blockchain development and other OSS development, we expect the differences in developers' perceptions of blockchain projects may affect their contribution dynamics and potentially alter previously identified relationships. Our study thus provides fresh empirical evidence about the effects of blockchain project context on the development dynamics and reevaluates existing theories of OSS development communities.

The first hypothesis that was tested concerns social diffusion within collaborative networks like OSS communities. Though prior literature has established the importance of peripheral developers' participation for project awareness and diffusion (Setia et al., 2012), the strong ideological motivations among blockchain developers (Bosu et al., 2019) may give core developers more influence over project popularity. We test the first hypothesis following Setia et al. (2012), which suggests the greater participation of peripheral developers in blockchain-based projects is associated with greater awareness about their product. Consistent with the prior literature, we measure awareness as the number of watches (Petryk et al., 2023). The result is presented in column (1) in Table 3.

Table 3. Hypothesis testing results: Effect on diffusion

Variable	(1) DV=Watches	(2) DV= Watches
Peripheral	0.515*** (0.0071)	0.487*** (0.053)
Core	-0.055*** (0.0107)	0.002 (0.005)
Superposition		14.967*** (3.727)
Superposition^2		-19.145*** (5.790)
Duration	0.0001** (0.00002)	-0.0004 (0.00004)
Repository FE	Y	Y
Num obs	101,081	1,593
Adj R-sq	0.833	0.756

Note: Fixed effects (FE) model is estimated; heteroskedasticity-consistent and autocorrelation-corrected standard errors in parentheses; + p<0.1, * p<0.05, ** p<0.01, *** p<0.001.

The result in column 1, Table 3 shows that the awareness of a repository increases when more developers show interest in it but are not yet heavily

involved in its development activity. A likely reason is that peripheral developers are involved in many other projects on the platform, therefore, when interaction with the focal repository is reflected on their profile, their collaborators are exposed to the focal project. The attention to the focal project can be reflected by subscribing to the repository updates (watching). We also use the number of forks as the measure of diffusion, and the results are consistent.

Blockchain OSS projects feature technological complexity, uncertain and rapidly changing objectives, and a high cost of defects (Bosu et al., 2019). These characteristics create an environment in which the autonomy of work may be impacted by the need for higher control. The possible effect of such differences is twofold: on the one hand, the known effects of the work autonomy may be suppressed by the need for more scrutinous code testing and review, and on the other hand, the fast-changing requirements may require higher autonomy. Together with the prior evidence of the relationship between project popularity and structure of contribution, we point attention to the study by Medappa and Srivastava (2019) that suggests a non-linear relationship between a project's popularity and its superposition. To measure superposition, we calculate the number of releases made over a period and divide it by the number of developers contributing to the repository within the same time frame.

The results in column 2 in Table 3 show a non-linear relationship between superposition and the number of watches (the coefficient for Superposition² is significant). As the level of superposition in the project increases, the project popularity increases but up to a certain point, after which further increases of superposition are accompanied by a decrease in project popularity (coefficient for Superposition² is negative). Hence, the developers are reluctant to follow such projects (Ryan & Deci, 2000). High superposition signifies a high level of work independence that satisfies the need for autonomy. However, the lack of collaboration and exchange of ideas creates a negative affective state and deters interest in the project (Medappa & Srivastava, 2019). Developers need a decent amount of challenge and autonomy to be interested in contributing to the project and opportunities to work with other developers on more complex tasks (Ke & Zhang, 2010).

The complexity and novelty of blockchain development are accompanied by large developer community involvement and the growth of a large network of repositories (subprojects) within the Ethereum OSS community (Hussey, 2021). Nevertheless, many subprojects under development

may create talent competition. Since the prior literature has pointed to the strategic value of network embeddedness for project success (Grewal et al., 2006), our third hypothesis will examine the association between network embeddedness and project success.

Subsequent to the extant literature, we consider a project's success as a project's rate of knowledge creation and measure it as the number of commits (Boh et al., 2007; Crowston et al., 2003) and include them as the main dependent variables in our analysis. The results are presented in Table 4.

Table 4. Hypothesis testing results: Effect on innovation

Variable	(1) DV= ln(Commits)	(2) DV= ln(Commits)
Degree	0.002* (0.001)	
Eigenvector	7.682*** (1.723)	
Betweenness	-0.0003 (0.0002)	
mcIntCohesion		-0.004 (0.005)
mcIntCohesion ²		0.0001 (0.0001)
Duration	-0.294*** (0.028)	-0.0001*** (0.00004)
Repository FE	Y	Y
Year FE	N	N
Num obs	1,634	101,080
Adj R-sq	0.113	0.012

Note: Fixed effects (FE) model is estimated; heteroskedasticity-consistent and autocorrelation-corrected standard errors in parentheses; + p<0.1, * p<0.05, ** p<0.01, *** p<0.001

The results in column 1, Table 4 signify that there is a statistically significant relationship between a network structure and project success. As the degree centrality increases, the projects become more interconnected with others by the common developers, which attracts more contributions (commits). As the eigenvector centrality increases, the projects become more central in the network, which turns them into magnets for contributions. We also use a number of forks as the alternative measure of innovation, and the results are consistent.

Furthering our exploration of the importance of network ties for OSS development, we test the role of repeated collaborations for project success. Due to the vast number of repositories within the Ethereum community and the limited developer's capacity to focus on projects, establishing repeated collaborations with other developers may be different than in other OSS projects. We test the impact of developer repeated collaboration on project success in our next hypothesis drawn from Singh et al. (2011). In particular, we test whether a moderate level of internal cohesion within a project is better for a project's success than very high or very low levels of internal cohesion.

The results in column 2, Table 4 reveal that there is no statistically significant relationship between internal cohesion and the number of commits (neither linear nor non-linear). One possible reason for our result is that coordination from the non-profit foundation makes the internal cohesion mechanism redundant as other motivations are present to commit. Nevertheless, to fully answer this question, we need to include more detailed observations on other non-profit-coordinated and token-funded OSS projects. We discuss this and other limitations in the last section of our study.

5. Conclusions and discussion

In this study, we empirically analyze the OSS development under different governance and funding models. External governance in the form of non-profits and the funding model in the form of publicly tradeable tokens create heterogeneity among OSS projects and potentially serve as additional coordinating mechanisms in addition to the digital technology governance of version control (Shaikh & Henfridsson, 2017).

Using the examples of blockchain-based and non-blockchain-based OSS projects, we try to distinguish similarities and differences with the other OSS projects and identify the role blockchain technology plays in the transformation of OSS development. First, we mark that the open-source model resonates with the ideas of transparency and decentralization that became the pillars of blockchain technology and its use cases (Lumineau et al., 2021). The open-source mode of development makes developers an important stakeholder group within the blockchain and Web3 ecosystem.

Second, we find a striking similarity between the organizational model of the Ethereum community and classical OSS projects, like Linux or Apache. The convergence of the classic projects to the non-profit governance model took years and decades, while Ethereum founders adopted the model from the beginning. Nevertheless, the deeper analysis reveals that the identified similarity also suggests that the social hierarchies of the non-blockchain OSS projects were inherited. Despite the decentralization claim of blockchain-based communities, the bulk of productive development activity is concentrated among the top 5% of developers. This may not pose an issue due to the other governance mechanisms in place, e.g., Ethereum Improvement Protocol voting (Fracassi et al., 2024). Our study suggests initial empirical evidence of the comparison between OSS development of blockchain technology that prioritizes

power decentralization in the network of agents and other technologies agnostic of this principle.

Our findings enrich the blockchain literature by establishing the evidence that blockchain software development inherits both decentralized and centralized elements of the OSS model. Centralization serves as an organizing element to establish ownership over software code and maintain it in the long term. Nevertheless, as the examples of the long-standing OSS projects illustrate, external governance, e.g., in the form of a non-profit foundation, is suggested to prevent conflicts among developers with concentration of power. In addition, the concentration of expertise and code ownership among the active minority exposes the project to the “bus problem,” when the risk of losing key technical experts could potentially lead to the project stagnation or threat to its existence (McLay, 1994; Avelino et al., 2016). Hence, counterintuitively, the role of external governance is to prevent the fragility of the OSS model due to its tendency to centralize.

More broadly, our findings contribute to the broader discussion of decentralized communities’ evolution and the suitable governance mode within such communities. The efficient governance mechanism should be tailored around the value that such communities create and coordinate the transactional flows for value creation. In the case of open-source software, the coordinated development effort creates the libre software, which is a non-competitive, non-rivalry public good. The public goods theory suggests that the non-profit model is theoretically the most efficient in this type of setting (Hansmann, 1980). This suggestion agrees with the theory of commons, which recognizes the possibility of external authority to help provide resources and facilitate self-organization efforts (Ostrom, 1990). Built on the principles of self-organization and collective action (Olson, 1971), the transformation of the OSS communities under the influence of the formal organizational and funding models serves as an excellent ground to evaluate their effects. Our study may help build a bridge in understanding self-organized communities creating public innovations in other domains, such as DAOs (Ellinger et al., 2024).

From a practical perspective, the identified centralization also bears some operational tradeoffs for external managers. On the one hand, if the manager aims to maintain the neutrality of the project in the long term, they want to account for the diversity of opinions sourced from a larger proportion of voluntary contributors. On the other hand, 1% or 5% of the larger community of developers is sufficient for the work planned and bears a smaller coordination load (extra effort required to coordinate the activity in a larger

community). Therefore, the concentration of effort in the blockchain OSS data signifies the tradeoffs that the centralized manager, such as a non-profit foundation, makes to meet the long-term development goals and satisfy the decentralization requirement. Our analysis allows practitioners to better manage their development communities.

Nevertheless, our study has limitations and potential directions for further extension. Primarily, we focus on the most prominent OSS communities, e.g., Ethereum, with many contributors and contributions. Nevertheless, to provide evidence for the generalizability of our results, we need to incorporate other OSS communities in the analysis, both blockchain and non-blockchain, with non-profit and without, and with token incentives and without. In addition, future research must look at the effects of hierarchization and the monetary benefits of the OSS development on the participation levels. Business managers should understand better whether sponsoring certain types of software development activities does not undermine the theorized learning processes and contribution. Moreover, the hired labor-volunteer collaboration inside the OSS communities should be further explored as a novel tendency in the OSS communities. Lastly, the definition of the long-term sustainability of the non-profit-governed and token-incentivized OSS model should be explored in future studies.

6. References

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